



RECOVERY & REPAIR

KPV

KPV – 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	L-Lysyl-L-prolyl-L-valine
SYNONYMS / ALIASES	KPV; alpha-MSH(11-13); lysine-proline-valine tripeptide
CAS NUMBER	67727-97-3
MOLECULAR FORMULA	C16H30N4O4
MOLECULAR WEIGHT	342.44 g/mol
AMINO ACID SEQUENCE	KPV (3 residues; H-Lys-Pro-Val-OH); corresponds to the C-terminal tripeptide of alpha-melanocyte-stimulating hormone (alpha-MSH)
SOURCE	Synthetic (solid-phase peptide synthesis)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Small linear tripeptide; generally stable. Avoid repeated freeze-thaw cycles.

STRUCTURAL NOTES

KPV is a small synthetic tripeptide composed of L-lysine, L-proline, and L-valine. It corresponds to the C-terminal sequence (residues 11-13) of alpha-melanocyte-stimulating hormone (alpha-MSH), which itself is a 13-amino-acid post-translational cleavage product of pro-opiomelanocortin (POMC). Despite its short length, KPV retains certain structural features of the parent alpha-MSH C-terminus and is widely used as a model alpha-MSH fragment in research.

SELECTED PEER-REVIEWED REFERENCES

1. Brzoska T. et al. - *Alpha-Melanocyte-stimulating hormone and related tripeptides: biochemistry, antiinflammatory and protective effects in vitro and in vivo, and future perspectives for the treatment of immune-mediated inflammatory diseases* - Endocrine Reviews, 2008 - DOI: 10.1210/er.2007-0027
2. Dalmasso G. et al. - *PepT1-mediated tripeptide KPV uptake reduces intestinal inflammation* - Gastroenterology, 2008 - DOI: 10.1053/j.gastro.2008.04.027
3. Land S.C. - *Inhibition of cellular and systemic inflammation cues in human bronchial epithelial cells by melanocortin-related peptides* - Peptides, 2012 - DOI: 10.1016/j.peptides.2011.10.025

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.